Password Manager

Python Internship Project

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Project category: Python Programming

Repository: upskillcampus

PROJECT REPORT

1. Project Overview

A command-line application to save and retrieve account credentials locally. It uses SQLite for storage and Fernet encryption from the cryptography package to encrypt passwords.

2. Objectives

The objective is to build a practical Python application that demonstrates core programming concepts and provides a simple, usable solution to the selected task.

3. Main Features

• Stores service name, username, and encrypted password in SQLite

• Generates a local encryption key on first run

• Can save and retrieve credentials by service name

• Uses parameterized SQL statements

4. Technologies Used

Python 3; SQLite (built-in); cryptography package (Fernet)

5. Program Workflow

1. Start the Python program.
2. Display the menu or request the required input.
3. Validate input and perform the selected operation.
4. Display the result and handle common errors.
5. Save or retrieve data where the project requires persistent storage.

6. Testing

The program should be tested with normal inputs, invalid inputs, empty fields, and repeated runs. For file operations, use a sample directory first. For database-backed projects, confirm that saved data remains available after restarting the program.

7. Challenges and Solutions

Typical challenges include validating user input, handling duplicate records or filenames, and keeping data available across sessions. These are addressed with conditional checks, unique identifiers or destination names, and local database storage where applicable.

8. Learning Outcomes

This project provides practice with Python functions, control flow, data structures, input validation, file handling, and/or database operations. It also develops debugging and project documentation skills.

9. Source Code and Repository

Repository name: upskillcampus

GitHub profile: https://github.com/divagar5943-lang

Repository URL: https://github.com/divagar5943-lang/upskillcampus

Note: Upload the matching .py file and this PDF to the repository before submitting. The repository and file links must be checked after upload; this report does not claim that the files have already been uploaded.

Security Note

This is an educational example, not a professionally audited password vault. Keep password.key private and backed up securely. Anyone who obtains both the database and key may decrypt stored passwords. Do not use it for real sensitive accounts without a security review.